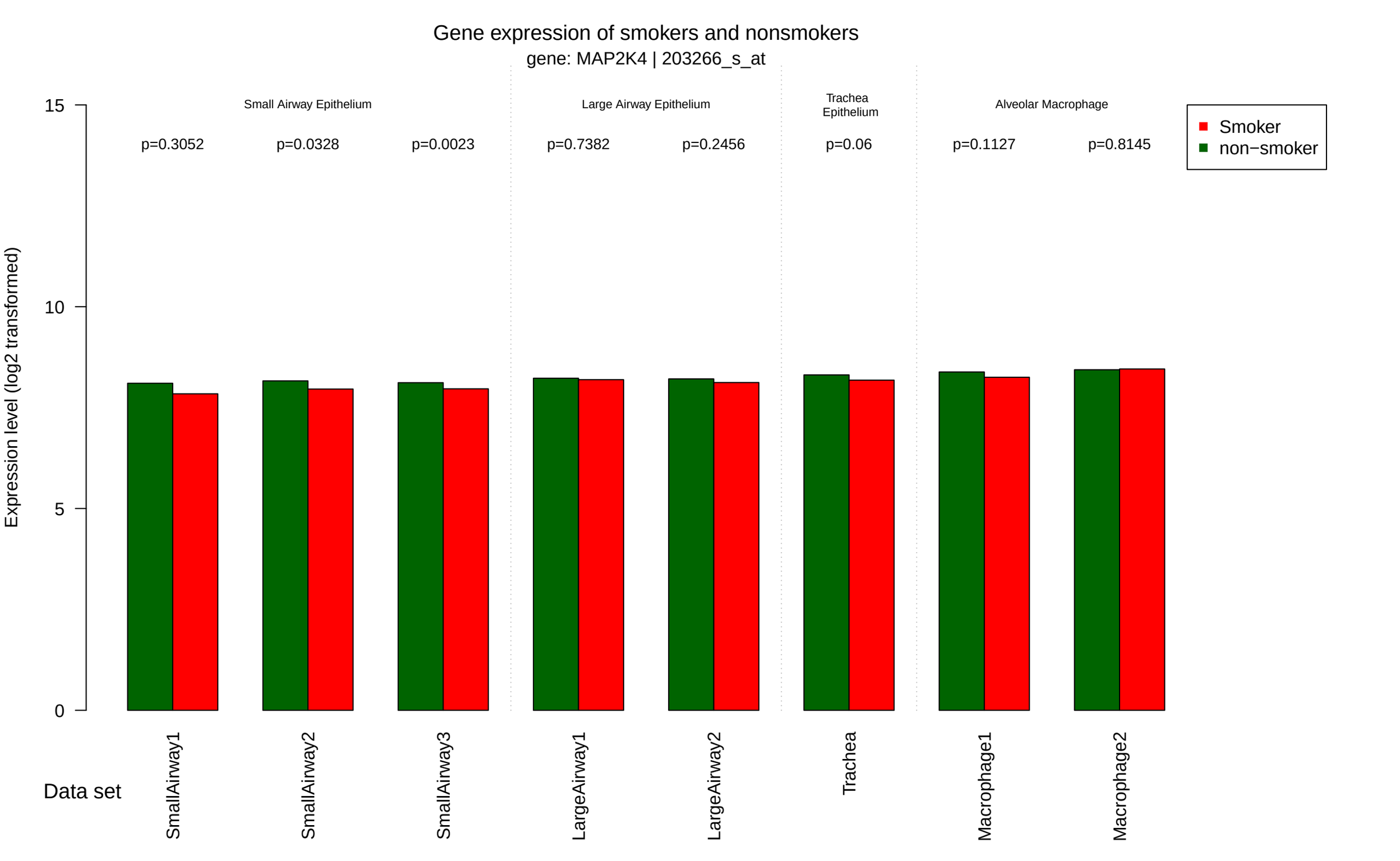
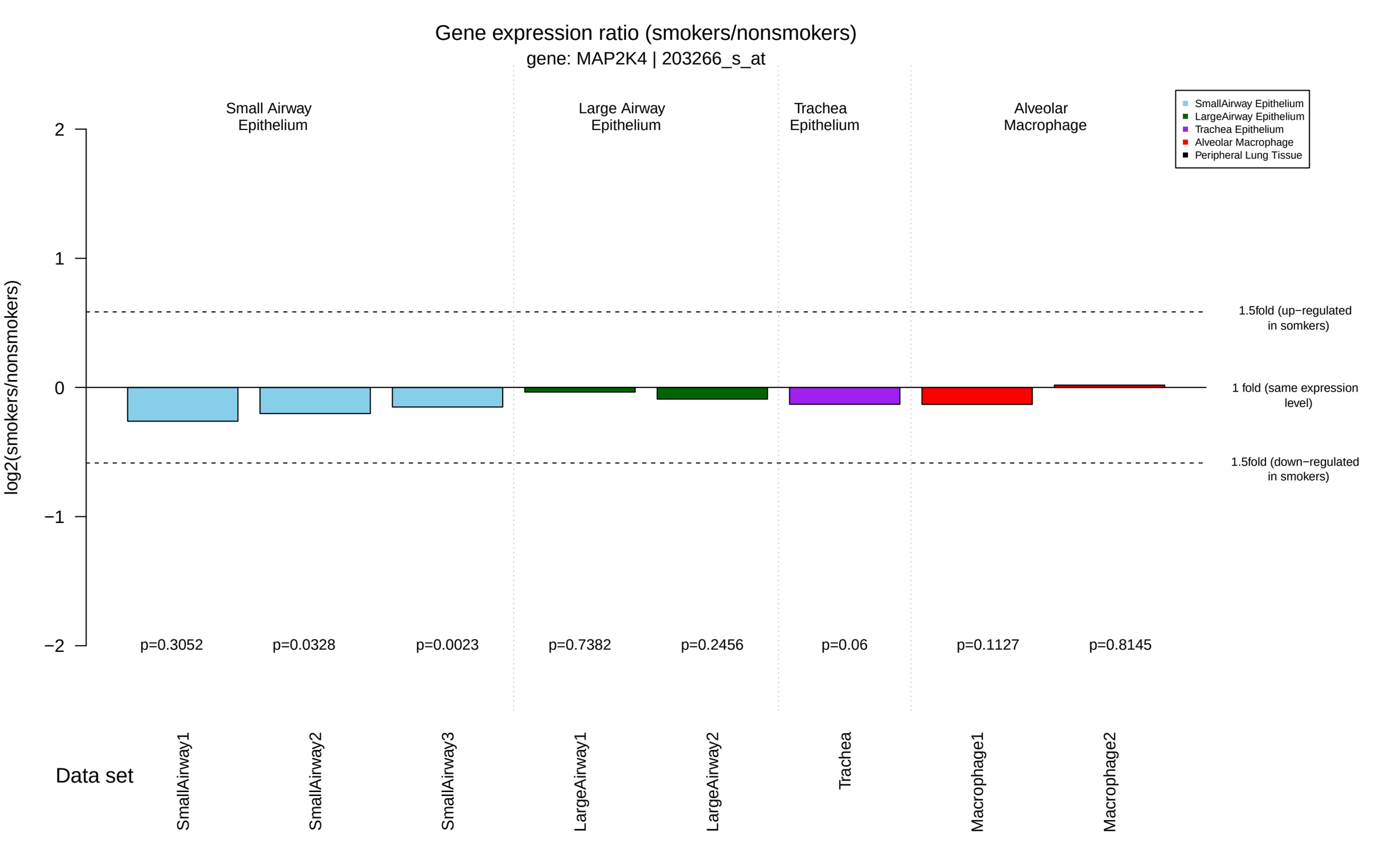
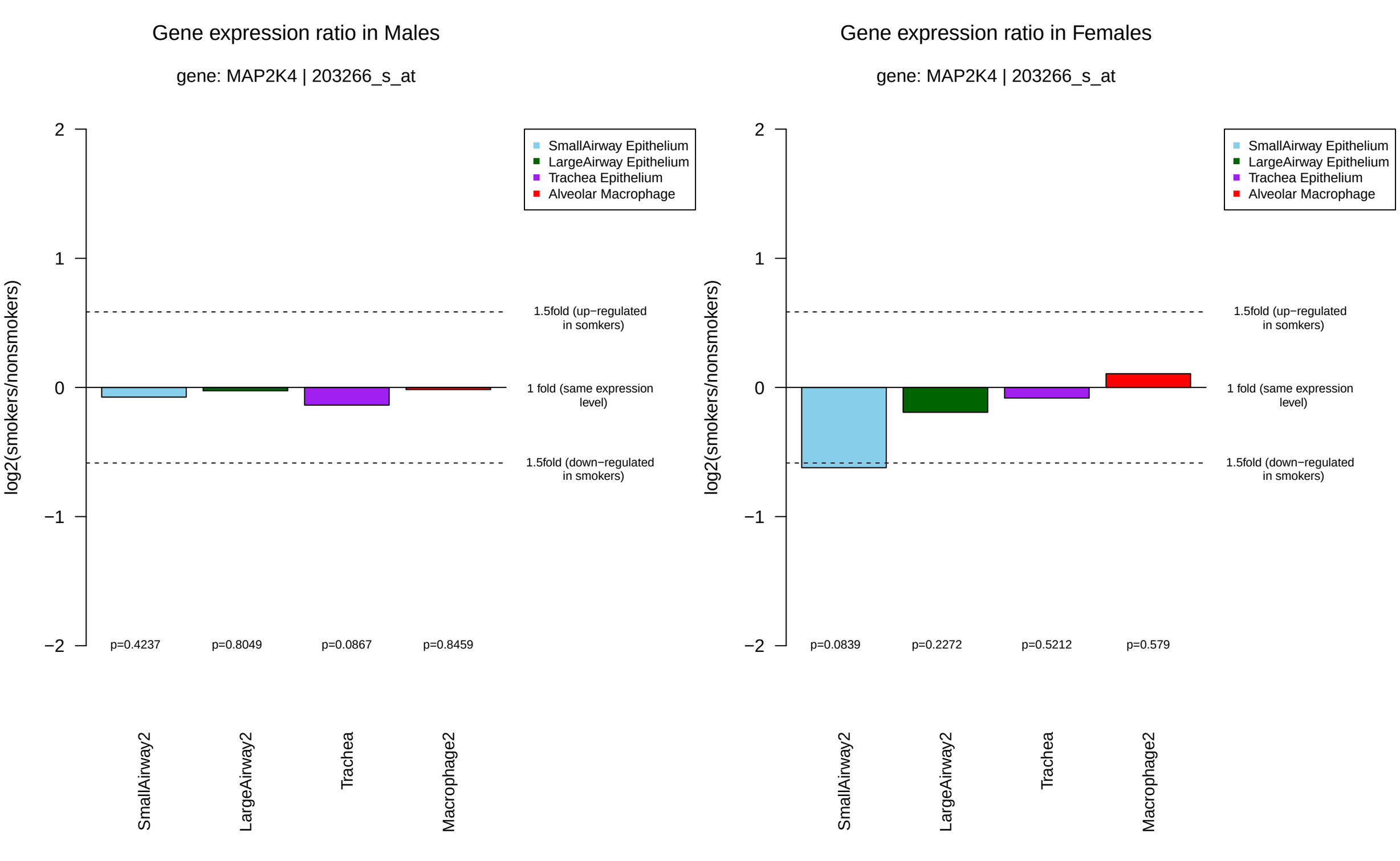
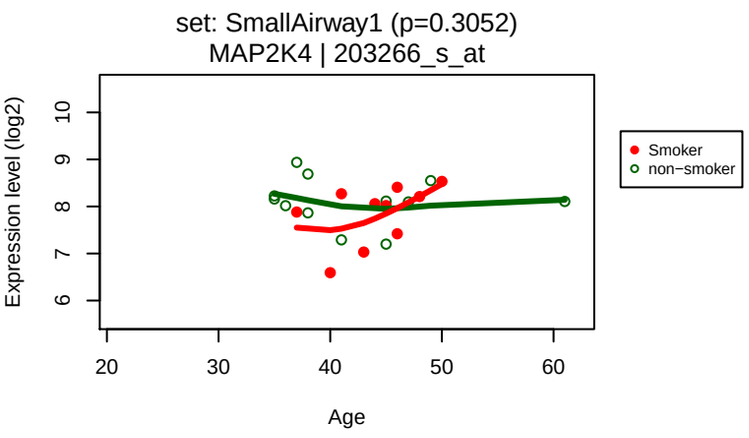
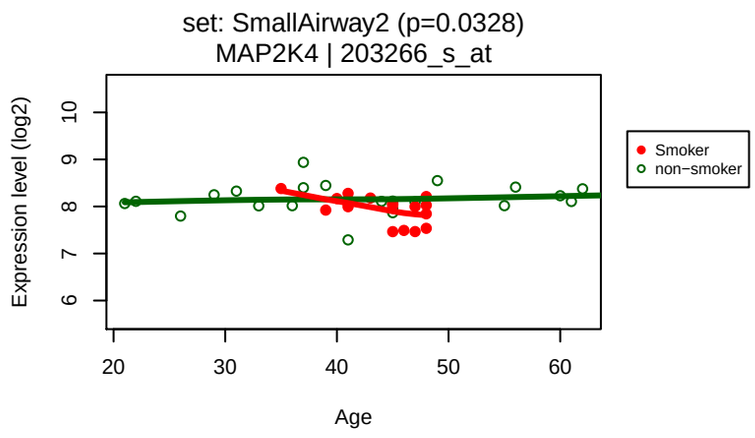
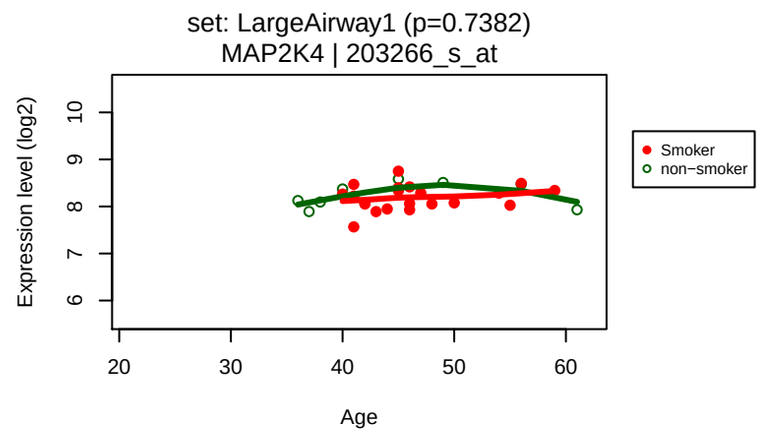

Gene expression of smokers and nonsmokers
gene: MAP2K4 | 203266_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3052
p=0.0328
p=0.0023
p=0.7382
p=0.2456
p=0.06
p=0.1127
p=0.8145
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

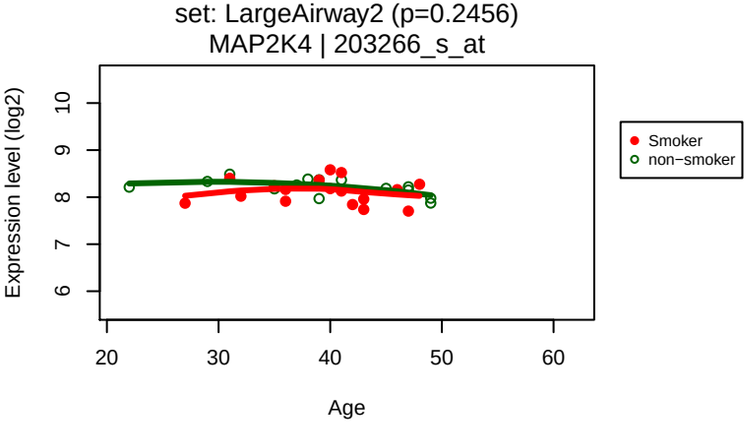

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K4 | 203266_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3052
p=0.0328
p=0.0023
p=0.7382
p=0.2456
p=0.06
p=0.1127
p=0.8145
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

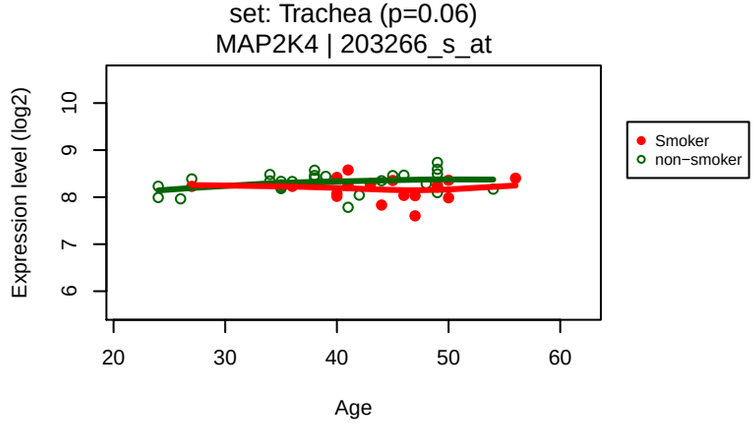

Gene expression ratio in Males
gene: MAP2K4 | 203266_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4237
p=0.8049
p=0.0867
p=0.8459
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K4 | 203266_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0839
p=0.2272
p=0.5212
p=0.579
SmallAirway2
LargeAirway2
Trachea
Macrophage2

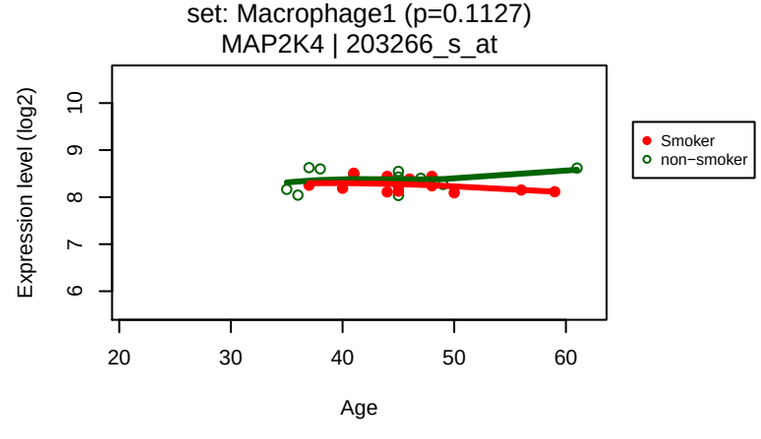

set: SmallAirway1 (p=0.3052)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker

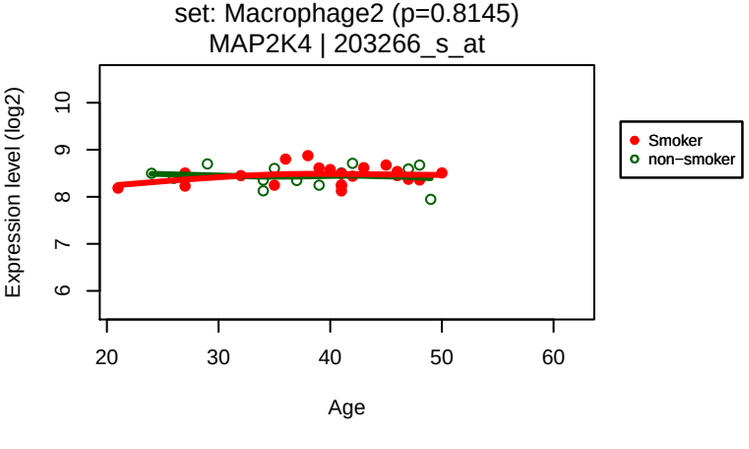

set: SmallAirway2 (p=0.0328)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker

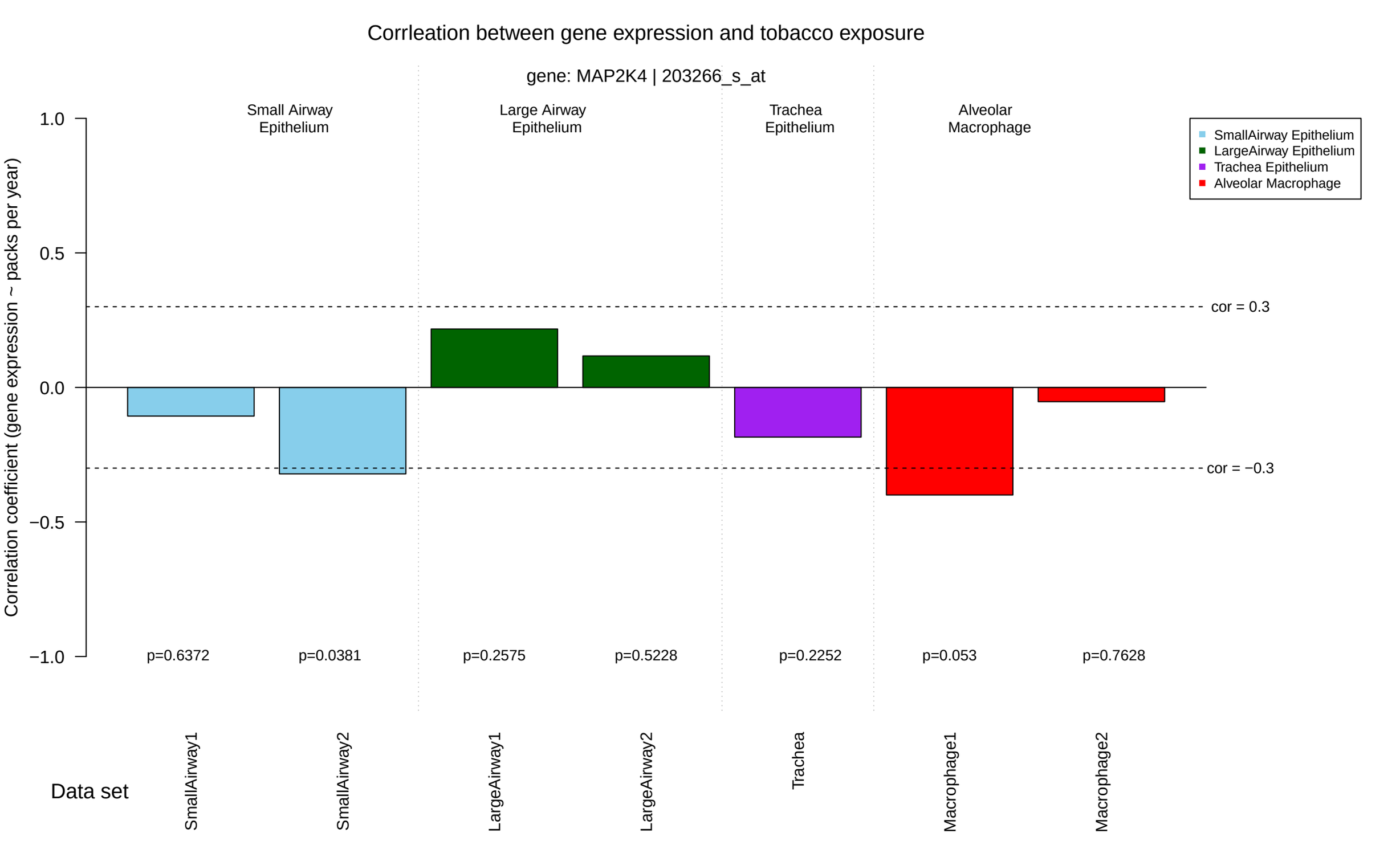

set: LargeAirway1 (p=0.7382)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2456)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.06)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1127)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8145)
MAP2K4 | 203266_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K4 | 203266_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6372
p=0.0381
p=0.2575
p=0.5228
p=0.2252
p=0.053
p=0.7628
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2